Predictive Modeling of Building Energy Efficiency

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1 & 2. Project Scope & Strategy

Bridging Design & Science

Sustainable architecture requires the ability to predict thermal behavior before construction. We leverage the UCI Energy Efficiency dataset to bridge architectural geometry with energy demand.

Key Target Variable: Heating Load (kWh/m²)

Goal: Determine if architectural parameters like height, surface area, and compactness can reliably forecast energy efficiency requirements.

3. Dataset: Variable Overview

Feature Group	Specific Variable	Description
Geometric	Relative Compactness	Measure of building shape efficiency.
Envelope	Surface Area / Roof Area	Total structural exposure measurements (m^2).
Massing	Overall Height	Binary split: 3.5m vs 7.0m structures.
Environmental	Orientation / Glazing	Building direction and window distribution.
Target	Heating Load	Energy required for heating (kWh/m^2).

Methodology: Data Cleaning

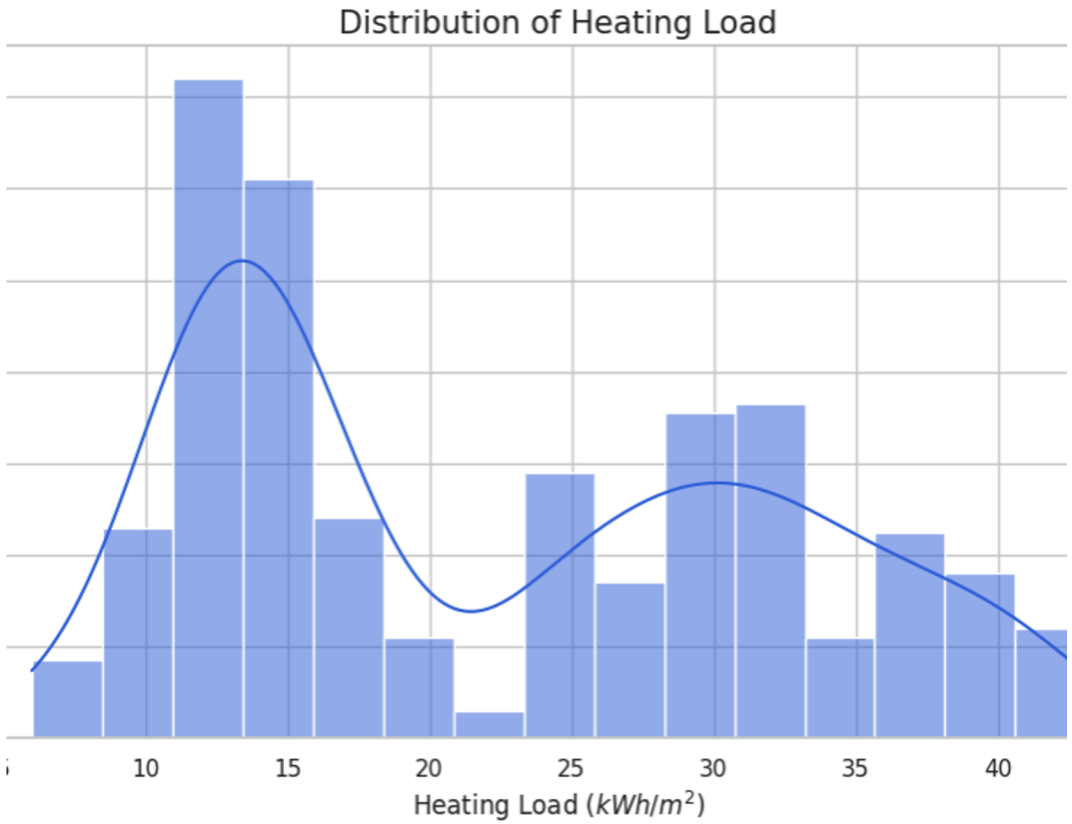
- **Relabeling:** Transformed generic "X1-Y2" indices into descriptive architectural features for interpretability.
- **CSV Cleanup:** Addressed trailing delimiters and ghost columns from spreadsheet exports to ensure matrix alignment.
- **Validation:** Verified 768 complete observations with zero missing values via df.info() check.
- **Splitting:** Identified "Overall Height" as the primary binary predictor for bimodal performance analysis.

4a. Heating Load Distribution

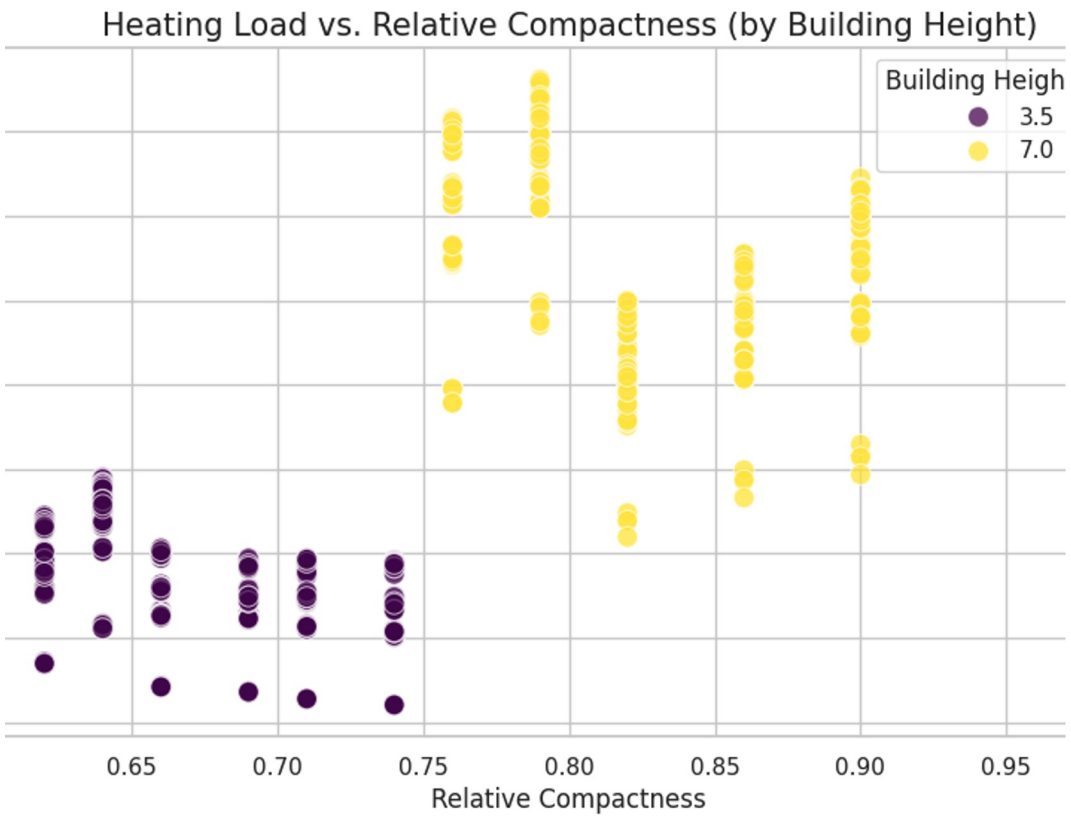
Bimodal Clusters

The histogram reveals two distinct performance clusters. This suggests a primary design factor creates a "split" in thermal behavior.

- **Efficient Designs:** 10–15 kWh/m²
- **High-Demand Designs:** 25–35 kWh/m²



Height: The Primary Driver



Height vs. Compactness

By color-coding the scatter plot by Height, we observe that vertical massing is the primary determinant of demand.

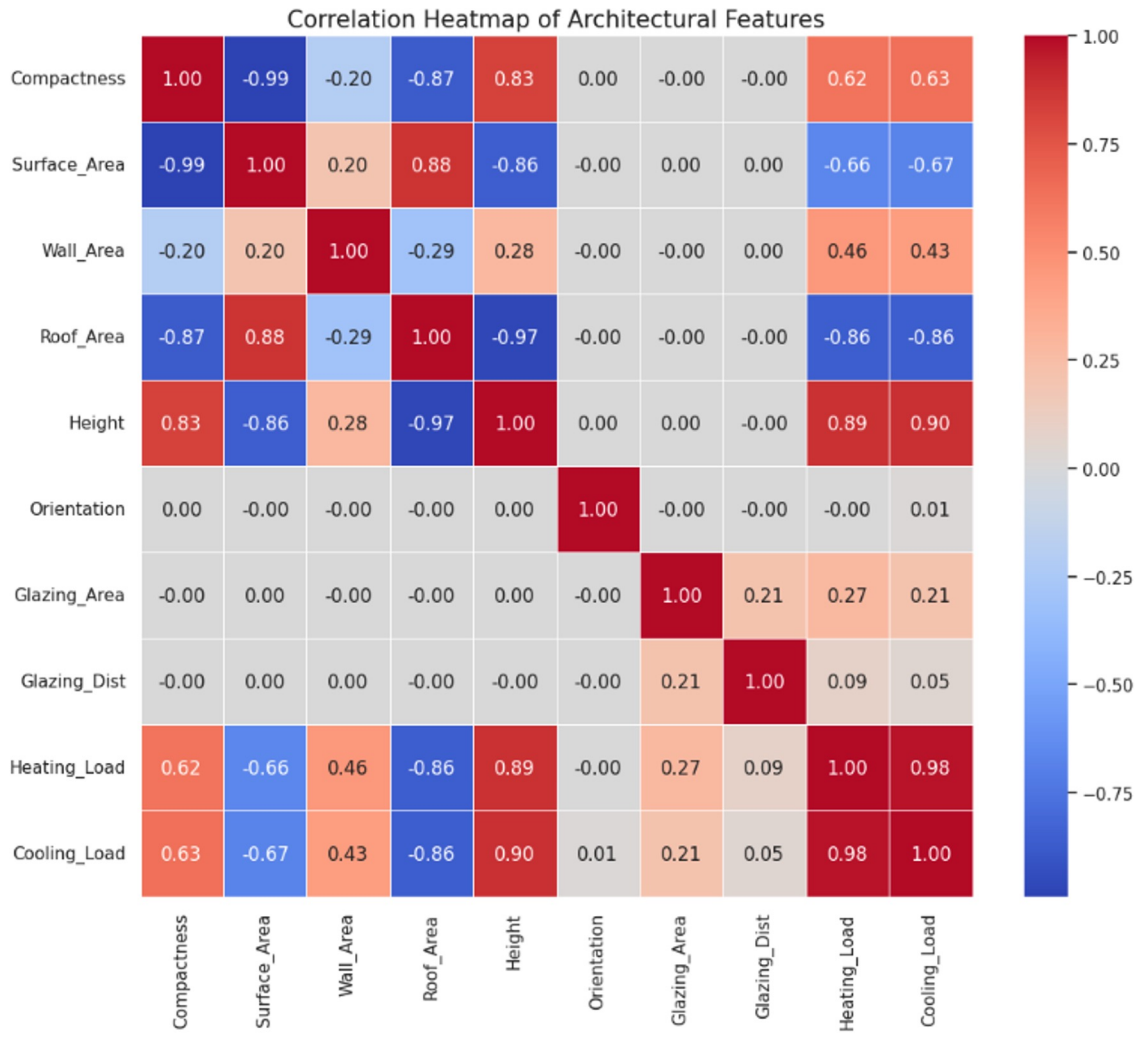
Regardless of compactness, 7.0m buildings consistently require more energy than 3.5m structures.

Numerical Evidence: Correlation

Feature Importance

The correlation matrix quantifies the physical drivers of energy efficiency:

0.89 correlation: Overall Height
0.62 correlation: Relative Compactness
0.00 correlation: Orientation (Negligible)



Modeling: Baseline vs. Improved

Baseline

Multiple Linear Regression assuming additive effects. Achieved 91% starting accuracy.

Reasoning

Thermal physics is non-linear. Interactions (e.g., Height x Surface Area) better reflect structural reality.

Improved Model

Polynomial Features (Degree 2) captured non-linear interactions, raising accuracy to 99%.

4b-e. Performance Benchmark

By introducing interaction terms, we reduced prediction error (RMSE) from 3.02 to 0.80 kWh/m².

4g. Final Predictive Validation

Testing the model on three unique design scenarios from the test set:

Design Sample	Actual Load	Predicted Load	Model Error
Building Design 1	16.47 kWh/m²	**16.56 kWh/m²**	0.09
Building Design 2	13.17 kWh/m²	**13.32 kWh/m²**	0.15
Building Design 3	32.82 kWh/m²	**32.79 kWh/m²**	0.03